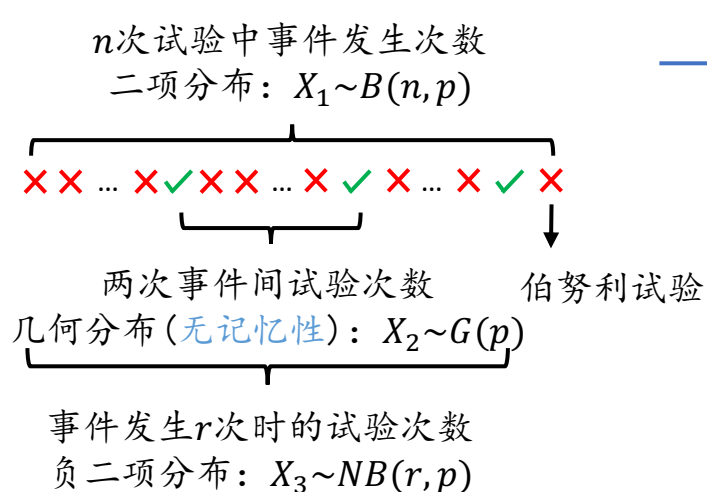


第二章 随机变量及其分布

离散



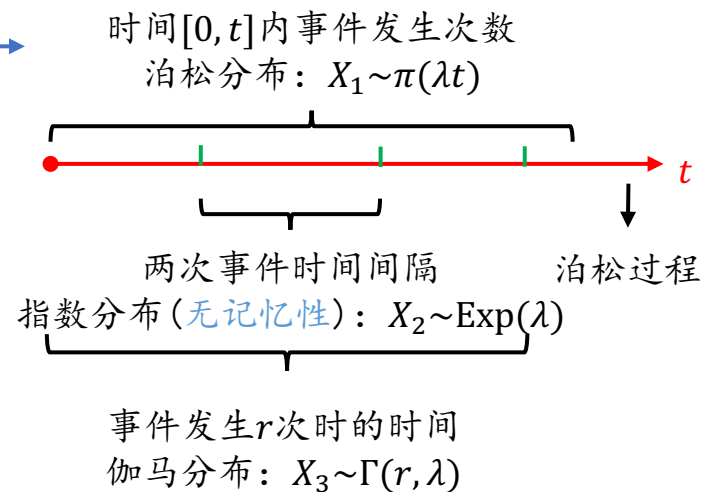
$$P(X_1 = k) = \binom{n}{k} p^k (1-p)^{n-k}, k = 0, 1, 2, \dots, n$$

$$P(X_2 = k) = (1-p)^{k-1} p, k = 1, 2, \dots$$

$$P(X_3 = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, k = r, r+1, \dots$$

$$n \rightarrow \infty, np_n = \lambda t$$

连续



$$P(X_1 = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, k = 0, 1, 2, \dots, n$$

$$f(x_2) = \begin{cases} \lambda e^{-\lambda x_2}, & x_2 > 0 \\ 0, & \text{else} \end{cases}$$

$$f(x_3) = \begin{cases} \frac{x_3^{r-1} \lambda^r e^{-\lambda x_3}}{\Gamma(r)}, & x_3 > 0 \\ 0, & \text{else} \end{cases}$$

正态分布 $X \sim N(\mu, \sigma^2)$: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

其他知识点

- 由泊松分布推导指数分布和伽马分布
- 随机变量的函数的分布